# Tarea2_Valenzuela_Ibañez

June 2, 2025

## 1 Tarea N°2

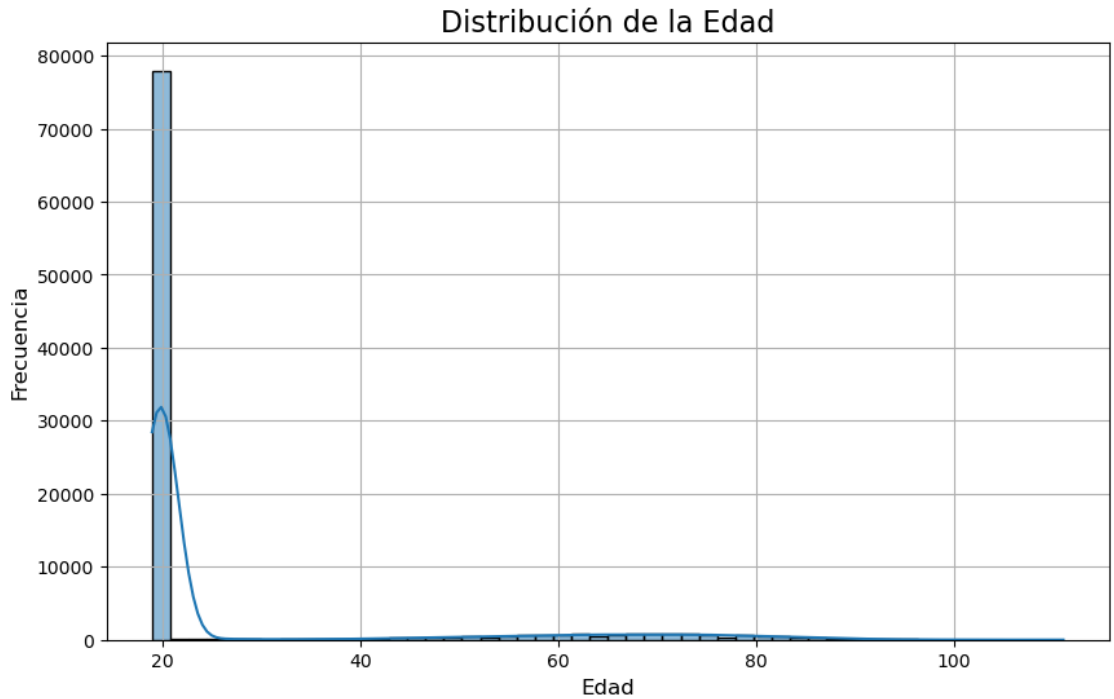
**Variable dictionary**

- folio_n20: identificacion individuo
- year: anio 20xx
- time: ronda de la encuesta (1-7)
- age: edad
- hombre: 1 si hombre, 0 mujer
- edu: educacion medida en anios completados
- region: region (1-16)
- status: 1 casado 2 convive 3 divorciado 4 viudo 5 soltero
- kids: 1 if has kids, 0 ow
- children: # of children under 18 years old, 18 included
- situation: 1 otro, 2 cesante, 3 trabajando, 4 inactivo
- lmp: 1 part-time worker, 2 full-time worker, 0 not working
- exp: years of labor experience
- size: tamanio de la firma
- occupation: codigo ocupacion
- wage: salario mensual en pesos del anio encuesta
- hours: horas semanales trabajadas
- informal: 1 if no work contract, 0 ow
- selfemp: 1 if self employed, 0 ow
- publicemp: 1 if worker in public sector, 0 ow
- sistema: 1 if in AFP system, 0 ow
- cotizando: 1 if contributor, 0 ow
- assets: accumulated wealth in pension system, pesos of each year
- exp_sist: years of experience in pension system
- expectancy: expectativa de vida autorreportada
- illness: 1 si tiene al menos una enferdad, 0 si no
- cronica: Si tiene enfermedad cronica 1, 0 eoc
- nocronica: Si tiene enfermedad no cronica 1, 0 eoc
- mental: Si tiene enfermedad mental 1, 0 eoc

1. Cargar la base de datos en el ambiente. Identifique los tipos de datos que se encuentran en la base, realice estadisticas descriptivas sobre las variables importantes (Hint: Revisar la distribuciones, datos faltantes, outliers, etc.) y limpie las variables cuando sea necesario.

```
          folio_n20  year  time  edad  hombre   edu  region  status  kids  \
0      124560204870     4     2    49     0.0   7.0     5.0     3.0     1
1      124560204870     6     3    51     0.0   4.0     5.0     3.0     1
2      124560204870     9     4    54     0.0   5.0     5.0     3.0     1
3      124560204870    12     5    57     0.0   4.0     NaN     5.0     0
4      124560205935     2     1  1982     0.0  15.0    13.0     3.0     1
…               …   …   …   …      …   …      …     …   …
96841  128133653188    12     5  1992     0.0  16.0     NaN     3.0     0
96842  128133692735    12     5  1992     0.0  12.0     NaN     2.0     0
96843  128133702320    12     5  1992     0.0  15.0     NaN     5.0     0
96844  128133728022    12     5  1992     0.0   8.0     NaN     2.0     0
96845  128133731927    12     5  1992     1.0   8.0     NaN     2.0     0

       children  …  fondob  fondoc  fondod  fondoe  exp_sist  expectancy  \
0           1.0  …     NaN     NaN     NaN     NaN         0        58.0
1           1.0  …     NaN     NaN     NaN     NaN         0        55.0
2           1.0  …     NaN     NaN     NaN     NaN         0        70.0
3           0.0  …     NaN     NaN     NaN     NaN         0        70.0
4           4.0  …     NaN     NaN     NaN     NaN        19        75.0
…         …  …     …     …     …     …       …          …
96841       0.0  …     NaN     NaN     NaN     NaN         7        80.0
96842       0.0  …     NaN     NaN     NaN     NaN         7        80.0
96843       0.0  …     NaN     NaN     NaN     NaN         7        75.0
96844       0.0  …     NaN     NaN     NaN     NaN         6        70.0
96845       0.0  …     NaN     NaN     NaN     NaN         7        90.0

       illness  cronica  nocronica  mental
0          1.0      0.0        1.0     0.0
1          0.0      0.0        0.0     0.0
2          1.0      0.0        0.0     1.0
3          1.0      0.0        0.0     0.0
4          NaN      NaN        NaN     NaN
…        …      …        …     …
96841      0.0      0.0        0.0     0.0
96842      1.0      0.0        0.0     0.0
96843      0.0      0.0        0.0     0.0
96844      1.0      0.0        0.0     0.0
96845      0.0      0.0        0.0     0.0

[96846 rows x 34 columns]

size                 70.010119
region                6.805650
cotizando             4.964583
edu                   4.053859
status                0.113582
hombre                0.051628
year                  0.000000
```

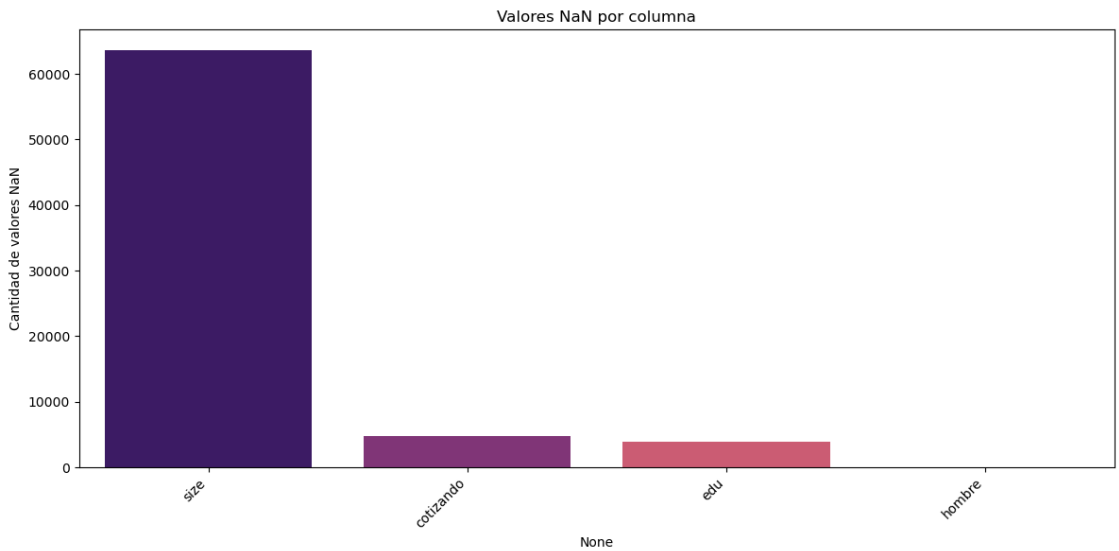
```
time                   0.000000
edad                   0.000000
kids                   0.000000
situation              0.000000
folio_n20              0.000000
lmp                    0.000000
exp                    0.000000
wage                   0.000000
hours                  0.000000
informal               0.000000
selfemp                0.000000
publicemp              0.000000
children               0.000000
sistema                0.000000
assets                 0.000000
exp_sist               0.000000
expectancy             0.000000
illness                0.000000
cronica                0.000000
nocronica              0.000000
mental                 0.000000
occupation_1.0         0.000000
occupation_2.0         0.000000
occupation_3.0         0.000000
occupation_4.0         0.000000
occupation_5.0         0.000000
occupation_6.0         0.000000
occupation_7.0         0.000000
occupation_8.0         0.000000
occupation_9.0         0.000000
occupation_10.0        0.000000
missing_informal       0.000000
missing_wage           0.000000
missing_hours          0.000000
missing_selfemp        0.000000
missing_publicemp      0.000000
missing_expectancy     0.000000
missing_sistema        0.000000
missing_children       0.000000
missing_illness        0.000000
missing_cronica        0.000000
missing_nocronica      0.000000
missing_mental         0.000000
missing_assets         0.000000
missing_exp            0.000000
dtype: float64
```




```
           folio_n20  year  time  hombre   edu  kids  children   exp  size  \
0       124560204870     4     2     0.0   7.0     1       1.0  12.0   NaN
1       124560204870     6     3     0.0   4.0     1       1.0  12.0   NaN
2       124560204870     9     4     0.0   5.0     1       1.0  12.0   NaN
3       124560204870    12     5     0.0   4.0     0       0.0  12.0   NaN
4       124560205935     2     1     0.0  15.0     1       4.0   6.0   NaN
…                …    …     …     …     …     …       …     …     …
96841   128133653188    12     5     0.0  16.0     0       0.0  12.0   NaN
96842   128133692735    12     5     0.0  12.0     0       0.0  12.0   NaN
96843   128133702320    12     5     0.0  15.0     0       0.0  12.0   NaN
96844   128133728022    12     5     0.0   8.0     0       0.0  12.0   NaN
96845   128133731927    12     5     1.0   8.0     0       0.0  12.0   NaN

            wage  …  region_6.0  region_7.0  region_8.0  region_9.0  \
0            0.0  …           0           0           0           0
1            0.0  …           0           0           0           0
2            0.0  …           0           0           0           0
3        50000.0  …           0           0           0           0
4            0.0  …           0           0           0           0
…          …   …           …           …           …           …
96841        0.0  …           0           0           0           0
96842   150000.0  …           0           0           0           0
96843        0.0  …           0           0           0           0
96844   183000.0  …           0           0           0           0
```

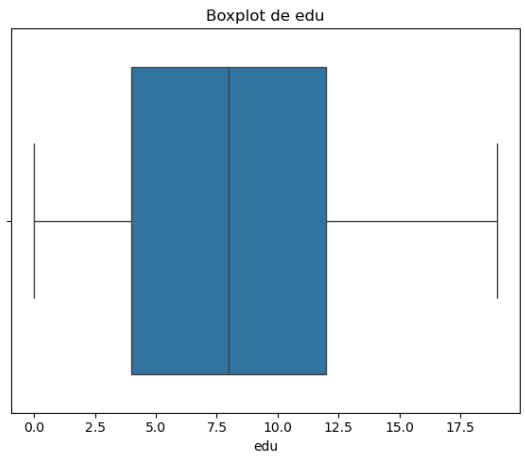
```
96845  270000.0  …           0           0           0           0

       region_10.0  region_11.0  region_12.0  region_13.0  region_14.0  \
0                0            0            0            0            0
1                0            0            0            0            0
2                0            0            0            0            0
3                0            0            0            0            0
4                0            0            0            1            0
…              …          …          …          …          …
96841            0            0            0            0            0
96842            0            0            0            0            0
96843            0            0            0            0            0
96844            0            0            0            0            0
96845            0            0            0            0            0

       region_15.0
0                0
1                0
2                0
3                0
4                0
…              …
96841            0
96842            0
96843            0
96844            0
96845            0

[92589 rows x 70 columns]

folio_n20        int64
year             int64
time             int64
hombre         float64
edu            float64
                 …
region_11.0      int64
region_12.0      int64
region_13.0      int64
region_14.0      int64
region_15.0      int64
Length: 70, dtype: object


Resumen por tipo:
int64      52
float64    18
Name: count, dtype: int64
```

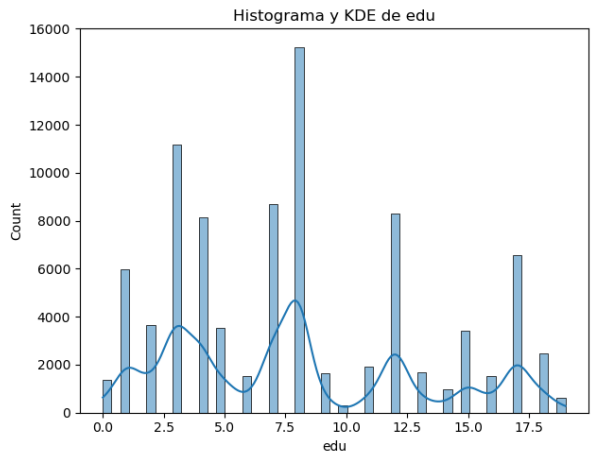
```
size         63565
cotizando     4793
edu           3866
hombre          48
year             0
time             0
children         0
exp              0
folio_n20        0
wage             0
dtype: int64
```

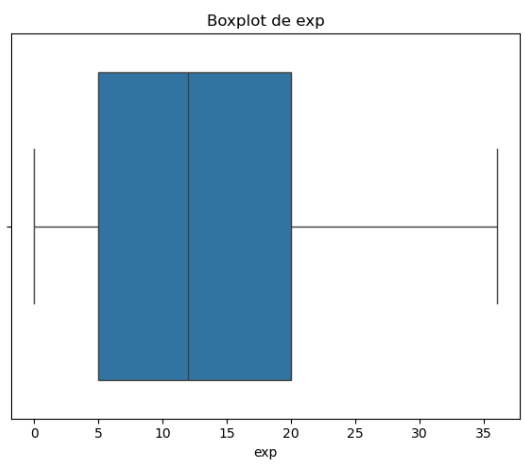


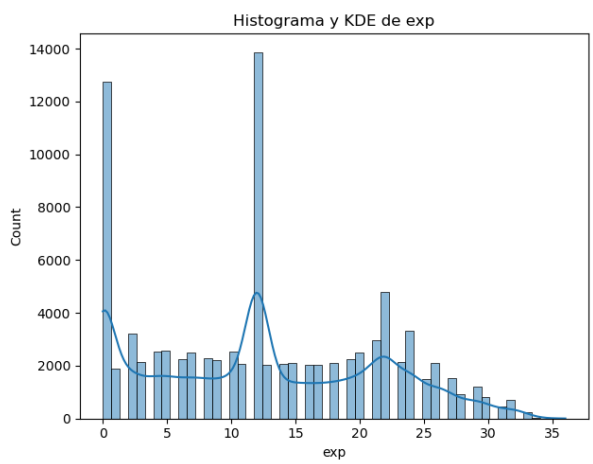

```
               count          mean           std           min           25%  \
folio_n20    92589.0  1.253244e+11  1.302004e+09  1.245602e+11  1.246218e+11
year         92589.0  7.946570e+00  4.663346e+00  2.000000e+00  4.000000e+00
time         92589.0  3.471395e+00  1.762499e+00  1.000000e+00  2.000000e+00
hombre       92541.0  4.956290e-01  4.999836e-01  0.000000e+00  0.000000e+00
edu          88723.0  7.969771e+00  5.111973e+00  0.000000e+00  4.000000e+00
…              …           …             …             …             …
region_11.0  92589.0  7.193079e-03  8.450690e-02  0.000000e+00  0.000000e+00
region_12.0  92589.0  8.845543e-03  9.363436e-02  0.000000e+00  0.000000e+00
region_13.0  92589.0  3.491235e-01  4.766957e-01  0.000000e+00  0.000000e+00
region_14.0  92589.0  6.793464e-03  8.214247e-02  0.000000e+00  0.000000e+00
region_15.0  92589.0  5.961831e-03  7.698280e-02  0.000000e+00  0.000000e+00

                     50%           75%           max
folio_n20    1.247077e+11  1.248372e+11  1.281337e+11
year         6.000000e+00  1.200000e+01  1.500000e+01
```

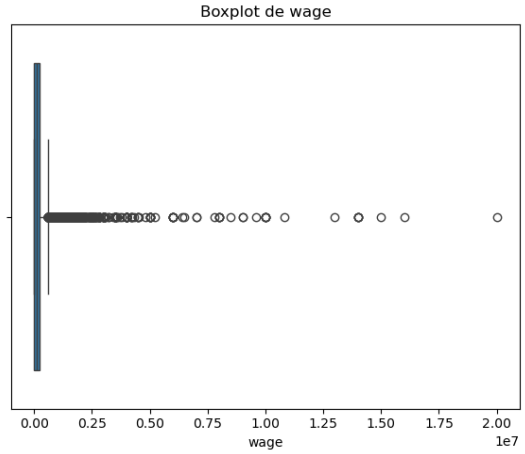
```
time          3.000000e+00  5.000000e+00  6.000000e+00
hombre        0.000000e+00  1.000000e+00  1.000000e+00
edu           8.000000e+00  1.200000e+01  1.900000e+01
…                      …             …             …
region_11.0   0.000000e+00  0.000000e+00  1.000000e+00
region_12.0   0.000000e+00  0.000000e+00  1.000000e+00
region_13.0   0.000000e+00  1.000000e+00  1.000000e+00
region_14.0   0.000000e+00  0.000000e+00  1.000000e+00
region_15.0   0.000000e+00  0.000000e+00  1.000000e+00

[69 rows x 8 columns]
```

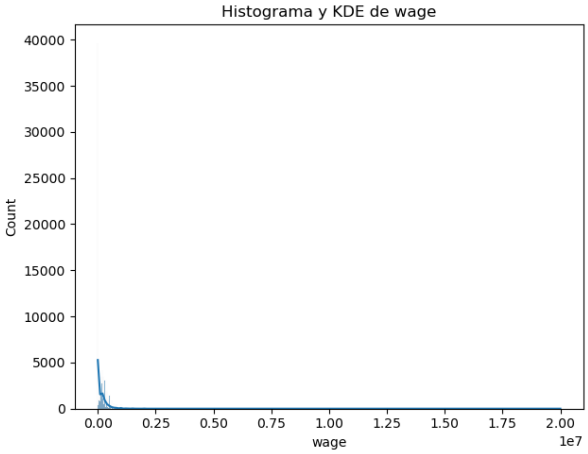


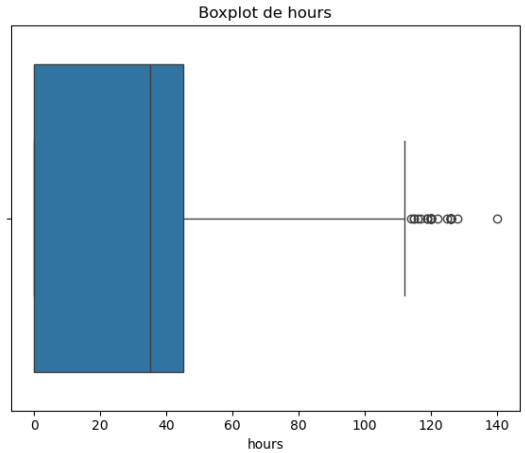



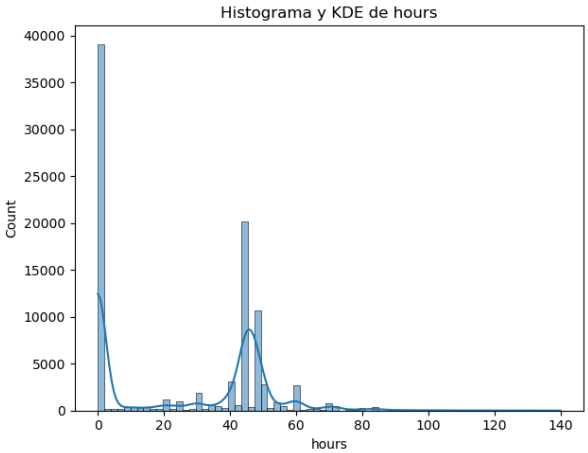



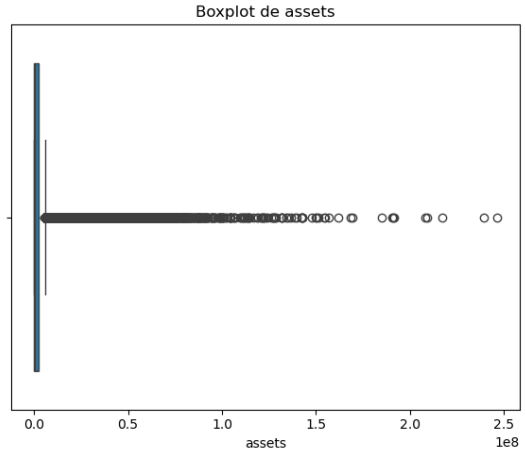



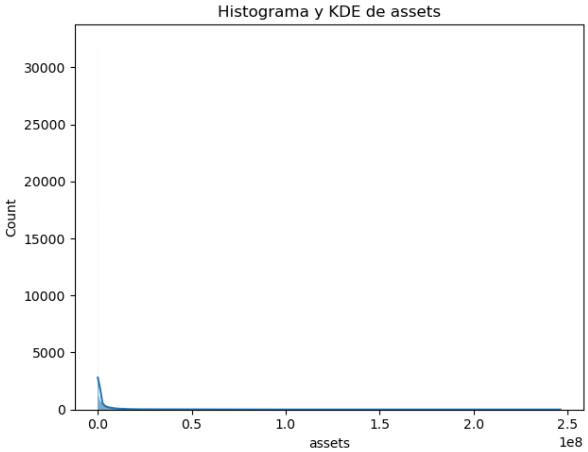

Boxplot de wage
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00
wage
1e7

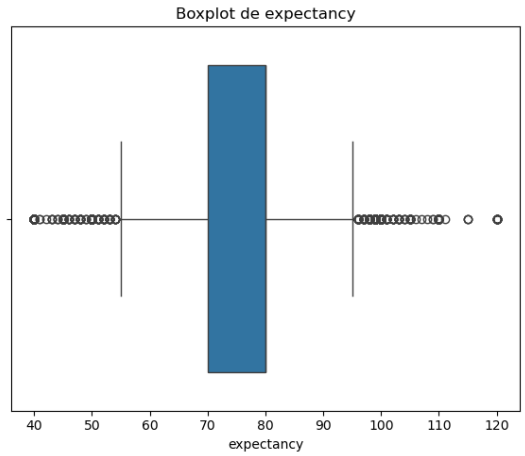

Histograma y KDE de wage
Count
40000
35000
30000
25000
20000
15000
10000
5000
0
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00
wage
1e7

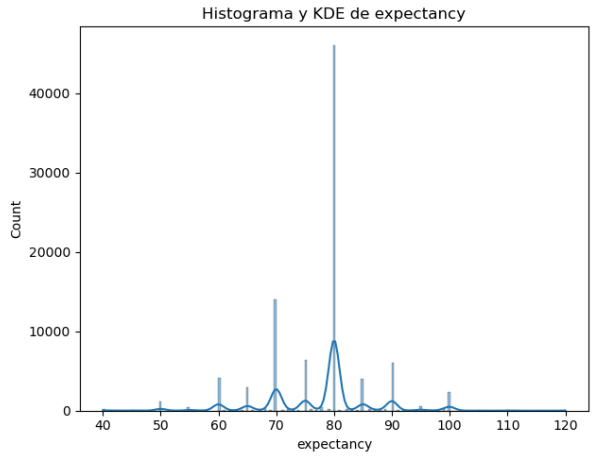

Boxplot de hours
0
20
40
60
80
100
120
140
hours


Histograma y KDE de hours
Count
40000
35000
30000
25000
20000
15000
10000
5000
0
0
20
40
60
80
100
120
140
hours


Boxplot de assets
0.0
0.5
1.0
1.5
2.0
2.5
assets
1e8


Histograma y KDE de assets
Count
30000
25000
20000
15000
10000
5000
0
0.0
0.5
1.0
1.5
2.0
2.5
assets
1e8




```
folio_n20: 18.70% outliers
cronica: 16.93% outliers
assets: 14.11% outliers
region_8.0: 11.83% outliers
selfemp: 10.98% outliers
status_2.0: 10.82% outliers
sistema: 10.37% outliers
occupation_9.0: 10.35% outliers
region_5.0: 9.83% outliers
publicemp: 8.67% outliers
occupation_7.0: 8.48% outliers
status_3.0: 8.47% outliers
situation_2: 8.33% outliers
occupation_5.0: 7.96% outliers
region_7.0: 6.57% outliers
informal: 6.41% outliers
status_4.0: 6.23% outliers
region_10.0: 6.16% outliers
mental: 6.04% outliers
nocronica: 5.88% outliers
children: 5.58% outliers
region_6.0: 5.55% outliers
occupation_4.0: 5.53% outliers
occupation_8.0: 5.44% outliers
region_9.0: 5.01% outliers
expectancy: 4.82% outliers
occupation_3.0: 4.57% outliers
region_4.0: 4.35% outliers
wage: 4.27% outliers
occupation_2.0: 4.09% outliers
lmp_1: 3.79% outliers
occupation_6.0: 2.79% outliers
```

```
region_2.0: 2.51% outliers
occupation_1.0: 2.08% outliers
region_3.0: 1.60% outliers
region_12.0: 0.88% outliers
region_11.0: 0.72% outliers
region_14.0: 0.68% outliers
region_15.0: 0.60% outliers
hours: 0.06% outliers
occupation_10.0: 0.05% outliers
year: 0.00% outliers
time: 0.00% outliers
hombre: 0.00% outliers
edu: 0.00% outliers
kids: 0.00% outliers
exp: 0.00% outliers
cotizando: 0.00% outliers
exp_sist: 0.00% outliers
illness: 0.00% outliers
status_5.0: 0.00% outliers
situation_3: 0.00% outliers
situation_4: 0.00% outliers
lmp_2: 0.00% outliers
region_13.0: 0.00% outliers
```

| | folio_n20 | year | time | hombre | edu | \ |
|---|---|---|---|---|---|---|
| count | 5.939100e+04 | 59391.000000 | 59391.000000 | 59353.000000 | 58505.000000 | |
| mean | 1.254674e+11 | 7.570019 | 3.429560 | 0.487743 | 7.796821 | |
| std | 1.403436e+09 | 3.146185 | 1.159956 | 0.499854 | 5.075466 | |
| min | 1.245602e+11 | 4.000000 | 2.000000 | 0.000000 | 0.000000 | |
| 25% | 1.246198e+11 | 4.000000 | 2.000000 | 0.000000 | 3.000000 | |
| 50% | 1.247935e+11 | 6.000000 | 3.000000 | 0.000000 | 7.000000 | |
| 75% | 1.248537e+11 | 12.000000 | 5.000000 | 1.000000 | 12.000000 | |
| max | 1.281337e+11 | 12.000000 | 5.000000 | 1.000000 | 19.000000 | |

| | kids | children | exp | wage | hours | \ |
|---|---|---|---|---|---|---|
| count | 59391.000000 | 59391.000000 | 59391.000000 | 5.939100e+04 | 59391.000000 | |
| mean | 0.363843 | 0.599434 | 12.683976 | 1.570034e+05 | 25.598794 | |
| std | 0.481108 | 0.947756 | 9.108814 | 3.254184e+05 | 24.353800 | |
| min | 0.000000 | 0.000000 | 0.000000 | 0.000000e+00 | 0.000000 | |

```
25%        0.000000      0.000000      4.000000  0.000000e+00      0.000000
50%        0.000000      0.000000     12.000000  1.000000e+05     30.000000
75%        1.000000      1.000000     21.000000  2.000000e+05     45.000000
max        1.000000      8.000000     33.000000  2.000000e+07    126.000000

       …    region_6.0    region_7.0    region_8.0    region_9.0  \
count  …  59391.000000  59391.000000  59391.000000  59391.000000
mean   …      0.054621      0.063865      0.116886      0.049789
std    …      0.227241      0.244514      0.321288      0.217510
min    …      0.000000      0.000000      0.000000      0.000000
25%    …      0.000000      0.000000      0.000000      0.000000
50%    …      0.000000      0.000000      0.000000      0.000000
75%    …      0.000000      0.000000      0.000000      0.000000
max    …      1.000000      1.000000      1.000000      1.000000

        region_10.0   region_11.0   region_12.0   region_13.0   region_14.0  \
count  59391.000000  59391.000000  59391.000000  59391.000000  59391.000000
mean       0.059622      0.006449      0.007998      0.330488      0.005843
std        0.236787      0.080046      0.089073      0.470393      0.076214
min        0.000000      0.000000      0.000000      0.000000      0.000000
25%        0.000000      0.000000      0.000000      0.000000      0.000000
50%        0.000000      0.000000      0.000000      0.000000      0.000000
75%        0.000000      0.000000      0.000000      1.000000      0.000000
max        1.000000      1.000000      1.000000      1.000000      1.000000

        region_15.0
count  59391.000000
mean       0.005338
std        0.072864
min        0.000000
25%        0.000000
50%        0.000000
75%        0.000000
max        1.000000

[8 rows x 69 columns]
```

## 2 Pooled OLS

2. Ejecute un modelo Pooled OLS para estimar la relacion entre los anios de educacion y la expectativa de vida. Seleccione las variables independientes a incluir en el modelo final e interprete su significado.

```
Resumen de multicolinealidad:
              Variable        VIF
41         situation_3        inf
```

| | | |
|---|---|---|
| 29 | missing_publicemp | inf |
| 28 | missing_selfemp | inf |
| 43 | lmp_1 | inf |
| 44 | lmp_2 | inf |
| 42 | situation_4 | 86.076592 |
| 5 | wage | 66.630662 |
| 40 | situation_2 | 33.478850 |
| 27 | missing_hours | 20.214818 |
| 26 | missing_wage | 12.695784 |
| 6 | hours | 10.305009 |
| 25 | missing_informal | 8.434994 |
| 56 | region_13.0 | 6.251521 |
| 34 | missing_assets | 4.621430 |
| 8 | selfemp | 3.938410 |
| 51 | region_8.0 | 3.503090 |
| 0 | year | 3.287589 |
| 48 | region_5.0 | 3.096248 |
| 11 | cotizando | 2.938728 |
| 23 | occupation_9.0 | 2.747175 |
| 50 | region_7.0 | 2.507709 |
| 53 | region_10.0 | 2.464340 |
| 21 | occupation_7.0 | 2.372733 |
| 19 | occupation_5.0 | 2.351576 |
| 49 | region_6.0 | 2.299587 |
| 12 | assets | 2.176963 |
| 52 | region_9.0 | 2.155470 |
| 47 | region_4.0 | 2.007133 |
| 18 | occupation_4.0 | 1.991197 |
| 35 | missing_exp | 1.924878 |
| 22 | occupation_8.0 | 1.924161 |
| 16 | occupation_2.0 | 1.921528 |
| 31 | missing_sistema | 1.853155 |
| 17 | occupation_3.0 | 1.725519 |
| 2 | edu | 1.698473 |
| 20 | occupation_6.0 | 1.617954 |
| 10 | sistema | 1.578722 |
| 1 | hombre | 1.570627 |
| 4 | exp | 1.541304 |
| 45 | region_2.0 | 1.535473 |
| 7 | informal | 1.504890 |
| 39 | status_5.0 | 1.488005 |
| 3 | kids | 1.481231 |
| 15 | occupation_1.0 | 1.399354 |
| 46 | region_3.0 | 1.346611 |
| 13 | cronica | 1.325396 |
| 32 | missing_children | 1.305700 |
| 9 | publicemp | 1.231606 |
| 55 | region_12.0 | 1.201558 |

```
38          status_4.0   1.184901
54         region_11.0   1.166993
36          status_2.0   1.136332
57         region_14.0   1.134621
58         region_15.0   1.123802
37          status_3.0   1.113060
30  missing_expectancy   1.085870
14              mental   1.079977
24     occupation_10.0   1.009061
33     missing_illness   1.004506
```

```
                          PanelOLS Estimation Summary
================================================================================
Dep. Variable:             expectancy   R-squared:                        0.0601
Estimator:                   PanelOLS   R-squared (Between):              0.0783
No. Observations:               54125   R-squared (Within):               0.0103
Date:                Tue, May 20 2025   R-squared (Overall):              0.0601
Time:                        20:53:54   Log-likelihood                -1.975e+05
Cov. Estimator:             Unadjusted
                                        F-statistic:                      64.063
Entities:                       22898   P-value                           0.0000
Avg Obs:                       2.3637   Distribution:                F(54,54070)
Min Obs:                       1.0000
Max Obs:                       4.0000   F-statistic (robust):             64.063
                                        P-value                           0.0000
Time periods:                       4   Distribution:                F(54,54070)
Avg Obs:                    1.353e+04
Min Obs:                    1.017e+04
Max Obs:                    1.654e+04

                                 Parameter Estimates
================================================================================
======
                    Parameter  Std. Err.     T-stat    P-value    Lower CI
Upper CI
--------------------------------------------------------------------------------
------
const                  78.523     0.6158     127.51     0.0000      77.316
79.730
year                   0.2090     0.0243     8.6061     0.0000      0.1614
0.2565
hombre                 1.4776     0.1002     14.741     0.0000      1.2811
1.6740
edu                    0.0268     0.0103     2.6136     0.0090      0.0067
0.0470
kids                  -1.5932     0.0999    -15.940     0.0000     -1.7891
-1.3973
exp                    0.0151     0.0055     2.7210     0.0065      0.0042
```

| | | | | | | |
|---|---|---|---|---|---|---|
| | | | | | | 0.0259 |
| wage | 0.0489 | 0.0449 | 1.0884 | 0.2764 | -0.0392 | 0.1370 |
| hours | -0.0043 | 0.0042 | -1.0226 | 0.3065 | -0.0127 | 0.0040 |
| informal | -0.2174 | 0.2006 | -1.0841 | 0.2783 | -0.6105 | 0.1757 |
| selfemp | -0.8429 | 0.2450 | -3.4406 | 0.0006 | -1.3231 | -0.3627 |
| publicemp | -0.3144 | 0.1820 | -1.7275 | 0.0841 | -0.6711 | 0.0423 |
| sistema | -2.5790 | 0.1668 | -15.461 | 0.0000 | -2.9059 | -2.2520 |
| cotizando | 1.0895 | 0.1391 | 7.8352 | 0.0000 | 0.8170 | 1.3621 |
| assets | -0.0151 | 0.0088 | -1.7108 | 0.0871 | -0.0324 | 0.0022 |
| cronica | -0.1136 | 0.1104 | -1.0284 | 0.3038 | -0.3300 | 0.1029 |
| mental | -2.0707 | 0.1612 | -12.849 | 0.0000 | -2.3866 | -1.7548 |
| occupation_1.0 | -0.3155 | 0.3396 | -0.9289 | 0.3530 | -0.9811 | 0.3502 |
| occupation_2.0 | 0.4407 | 0.2833 | 1.5554 | 0.1199 | -0.1146 | 0.9960 |
| occupation_3.0 | 0.3463 | 0.2646 | 1.3088 | 0.1906 | -0.1723 | 0.8649 |
| occupation_4.0 | -0.5484 | 0.2514 | -2.1812 | 0.0292 | -1.0412 | -0.0556 |
| occupation_5.0 | -1.2407 | 0.2277 | -5.4493 | 0.0000 | -1.6870 | -0.7945 |
| occupation_6.0 | -1.5952 | 0.2961 | -5.3873 | 0.0000 | -2.1756 | -1.0148 |
| occupation_7.0 | -1.2085 | 0.2209 | -5.4713 | 0.0000 | -1.6415 | -0.7756 |
| occupation_8.0 | -1.3018 | 0.2445 | -5.3232 | 0.0000 | -1.7811 | -0.8225 |
| occupation_9.0 | -1.3999 | 0.2139 | -6.5431 | 0.0000 | -1.8192 | -0.9805 |
| occupation_10.0 | 2.1459 | 1.4793 | 1.4506 | 0.1469 | -0.7535 | 5.0453 |
| missing_informal | 0.5892 | 0.2304 | 2.5574 | 0.0105 | 0.1376 | 1.0408 |
| missing_wage | 0.6421 | 0.2825 | 2.2732 | 0.0230 | 0.0885 | 1.1957 |
| missing_hours | 0.4589 | 0.3500 | 1.3111 | 0.1898 | -0.2271 | 1.1449 |
| missing_expectancy | 2.3775 | 0.1203 | 19.756 | 0.0000 | 2.1417 | |

| | | | | | | |
|---|---|---|---|---|---|---|
| 2.6134 | | | | | | |
| missing_sistema | 0.7676 | 0.1391 | 5.5177 | 0.0000 | 0.4949 | 1.0403 |
| missing_children | -0.5767 | 0.1350 | -4.2710 | 0.0000 | -0.8413 | -0.3120 |
| missing_illness | -0.1130 | 1.0642 | -0.1062 | 0.9154 | -2.1988 | 1.9727 |
| missing_assets | -0.8969 | 0.2124 | -4.2220 | 0.0000 | -1.3133 | -0.4805 |
| missing_exp | -0.7756 | 0.1825 | -4.2495 | 0.0000 | -1.1334 | -0.4179 |
| status_2.0 | -0.5459 | 0.1380 | -3.9566 | 0.0001 | -0.8163 | -0.2755 |
| status_3.0 | -0.8301 | 0.1502 | -5.5268 | 0.0000 | -1.1245 | -0.5357 |
| status_4.0 | 2.7423 | 0.1826 | 15.020 | 0.0000 | 2.3844 | 3.1001 |
| status_5.0 | -0.5700 | 0.1111 | -5.1318 | 0.0000 | -0.7877 | -0.3523 |
| situation_2 | -2.3241 | 0.5912 | -3.9314 | 0.0001 | -3.4828 | -1.1654 |
| situation_4 | -1.0262 | 0.5770 | -1.7785 | 0.0753 | -2.1570 | 0.1047 |
| region_2.0 | -3.2272 | 0.3303 | -9.7713 | 0.0000 | -3.8745 | -2.5798 |
| region_3.0 | -1.2084 | 0.3887 | -3.1090 | 0.0019 | -1.9702 | -0.4466 |
| region_4.0 | -0.3149 | 0.2783 | -1.1315 | 0.2579 | -0.8604 | 0.2306 |
| region_5.0 | -0.4834 | 0.2364 | -2.0447 | 0.0409 | -0.9468 | -0.0200 |
| region_6.0 | -0.2248 | 0.2633 | -0.8539 | 0.3932 | -0.7410 | 0.2913 |
| region_7.0 | -0.9736 | 0.2529 | -3.8504 | 0.0001 | -1.4693 | -0.4780 |
| region_8.0 | -1.6951 | 0.2298 | -7.3751 | 0.0000 | -2.1456 | -1.2446 |
| region_9.0 | -1.1376 | 0.2657 | -4.2819 | 0.0000 | -1.6584 | -0.6169 |
| region_10.0 | -0.9266 | 0.2590 | -3.5777 | 0.0003 | -1.4342 | -0.4190 |
| region_11.0 | -1.8225 | 0.5275 | -3.4548 | 0.0006 | -2.8564 | -0.7885 |
| region_12.0 | -1.0396 | 0.4815 | -2.1591 | 0.0308 | -1.9834 | -0.0959 |
| region_13.0 | -0.4745 | 0.2109 | -2.2494 | 0.0245 | -0.8879 | -0.0610 |
| region_14.0 | -0.9405 | 0.5695 | -1.6515 | 0.0987 | -2.0566 | |

```
0.1757
region_15.0           -2.4285     0.5831    -4.1650     0.0000     -3.5714
-1.2857
==============================================================================
======
```

R: Primero podemos notar que tenemos un R cuadrado Overall con un valor de 0.0628, es decir, que las variables independientes solo pueden explicar el 6,28% de la variación en la expectativa de vida. Podemos suponer que la expectativa de vida esta relacionada a variables que no poseemos, en función de esto sugeriría buscar más datos relacionados a la salud y agregarlos al data frame.

Luego, tenemos que la constante es de 79,8 años correspondiente a la expectativa de vida base, manteniendo el resto de variables como 0. En cuanto a la relación de los años de eduación con la expectativa de vida, el modelo Pooled OLS nos entrega un coeficiente negativo de 0,028, lo que significa que por cada año de educación, la expectativa de vida baja en 0,03 años aproximadamente. Además, esta variable cuenta con un p muy alto, por lo que no es significativa.

Otras variables relevantes son year con un coeficiente de 0,1437, que significa que por cada año, nuestra expectativa de vida aumenta en 0,14 años, y también vemos que los hombres viven 1,4 años más que las mujeres aproximadamente.

Finalmente, a modo general notamos que hay varias variables con un p alto, por lo que no son significativas y otras con un VIF muy alto lo que implica una gran colinealidad.

# 3 Efectos Fijos

3. Ejecute un modelo efectos fijos para estimar la misma relacion anterior. Seleccione las variables independientes a incluir en el modelo final e interprete su significado.

```
                          PanelOLS Estimation Summary
================================================================================
Dep. Variable:             expectancy   R-squared:                        0.0137
Estimator:                   PanelOLS   R-squared (Between):             -0.0288
No. Observations:               54125   R-squared (Within):               0.0137
Date:                Tue, May 20 2025   R-squared (Overall):             -0.0123
Time:                        20:53:55   Log-likelihood                -1.744e+05
Cov. Estimator:                Robust
                                        F-statistic:                      16.649
Entities:                       22898   P-value                           0.0000
Avg Obs:                       2.3637   Distribution:                F(26,31201)
Min Obs:                       1.0000
Max Obs:                       4.0000   F-statistic (robust):             13.170
                                        P-value                           0.0000
Time periods:                       4   Distribution:                F(26,31201)
Avg Obs:                   1.353e+04
Min Obs:                   1.017e+04
Max Obs:                   1.654e+04
```

Parameter Estimates

| | Parameter | Std. Err. | T-stat | P-value | Lower CI | Upper CI |
|---|---|---|---|---|---|---|
| const | 75.010 | 0.9878 | 75.933 | 0.0000 | 73.074 | 76.946 |
| edu | -0.0219 | 0.0258 | -0.8476 | 0.3967 | -0.0725 | 0.0287 |
| kids | -0.2747 | 0.2429 | -1.1311 | 0.2580 | -0.7508 | 0.2013 |
| exp | 0.0885 | 0.0375 | 2.3588 | 0.0183 | 0.0150 | 0.1620 |
| cronica | -0.6176 | 0.1604 | -3.8492 | 0.0001 | -0.9321 | -0.3031 |
| sistema | 0.1199 | 0.2163 | 0.5546 | 0.5791 | -0.3039 | 0.5438 |
| wage | 0.0476 | 0.0207 | 2.3031 | 0.0213 | 0.0071 | 0.0882 |
| missing_assets | -0.1183 | 0.2140 | -0.5527 | 0.5805 | -0.5376 | 0.3011 |
| status_4.0 | -0.6103 | 0.3923 | -1.5554 | 0.1198 | -1.3792 | 0.1587 |
| informal | -0.5122 | 0.2472 | -2.0718 | 0.0383 | -0.9968 | -0.0276 |
| region_3.0 | 1.8588 | 3.4957 | 0.5317 | 0.5949 | -4.9929 | 8.7104 |
| region_4.0 | 0.3352 | 1.7162 | 0.1953 | 0.8452 | -3.0287 | 3.6991 |
| region_5.0 | 0.4408 | 1.5019 | 0.2935 | 0.7691 | -2.5030 | 3.3846 |
| region_6.0 | -0.8094 | 1.5859 | -0.5104 | 0.6098 | -3.9178 | 2.2990 |
| region_8.0 | -0.9193 | 1.9284 | -0.4767 | 0.6336 | -4.6991 | 2.8605 |
| region_9.0 | -1.8715 | 1.8098 | -1.0341 | 0.3011 | -5.4188 | 1.6759 |
| region_11.0 | 2.8322 | 4.9068 | 0.5772 | 0.5638 | -6.7853 | 12.450 |
| region_13.0 | 1.2083 | 1.2039 | 1.0036 | 0.3156 | -1.1515 | 3.5681 |
| year | 0.1944 | 0.0316 | 6.1612 | 0.0000 | 0.1326 | 0.2563 |
| situation_4 | -0.2233 | 0.2113 | -1.0568 | 0.2906 | -0.6375 | 0.1909 |
| occupation_2.0 | 0.2025 | 0.3453 | 0.5865 | 0.5575 | -0.4743 | |

```
0.8794
occupation_3.0     0.4502     0.2925     1.5390     0.1238     -0.1232
1.0236
occupation_5.0    -0.4727     0.2630    -1.7976     0.0722     -0.9882
0.0427
occupation_6.0    -1.2303     0.3871    -3.1783     0.0015     -1.9890
-0.4716
occupation_7.0    -0.3511     0.2774    -1.2654     0.2057     -0.8949
0.1927
occupation_8.0    -0.7188     0.3253    -2.2097     0.0271     -1.3565
-0.0812
occupation_9.0    -0.4083     0.2570    -1.5883     0.1122     -0.9121
0.0955
================================================================================
==

F-test for Poolability: 1.9165
P-value: 0.0000
Distribution: F(22897,31201)

Included effects: Entity

Resumen de multicolinealidad:
              Variable        VIF
41         situation_3        inf
29   missing_publicemp        inf
28     missing_selfemp        inf
43               lmp_1        inf
44               lmp_2        inf
42         situation_4  86.076592
5                 wage  66.630662
40         situation_2  33.478850
27       missing_hours  20.214818
26        missing_wage  12.695784
6                hours  10.305009
25    missing_informal   8.434994
56         region_13.0   6.251521
34      missing_assets   4.621430
8               selfemp   3.938410
51          region_8.0   3.503090
0                 year   3.287589
48          region_5.0   3.096248
11           cotizando   2.938728
23      occupation_9.0   2.747175
50          region_7.0   2.507709
53         region_10.0   2.464340
21      occupation_7.0   2.372733
19      occupation_5.0   2.351576
```

```
49          region_6.0   2.299587
12              assets   2.176963
52          region_9.0   2.155470
47          region_4.0   2.007133
18      occupation_4.0   1.991197
35         missing_exp   1.924878
22      occupation_8.0   1.924161
16      occupation_2.0   1.921528
31     missing_sistema   1.853155
17      occupation_3.0   1.725519
2                  edu   1.698473
20      occupation_6.0   1.617954
10             sistema   1.578722
1               hombre   1.570627
4                  exp   1.541304
45          region_2.0   1.535473
7             informal   1.504890
39          status_5.0   1.488005
3                 kids   1.481231
15      occupation_1.0   1.399354
46          region_3.0   1.346611
13             cronica   1.325396
32    missing_children   1.305700
9            publicemp   1.231606
55         region_12.0   1.201558
38          status_4.0   1.184901
54         region_11.0   1.166993
36          status_2.0   1.136332
57         region_14.0   1.134621
58         region_15.0   1.123802
37          status_3.0   1.113060
30  missing_expectancy   1.085870
14              mental   1.079977
24     occupation_10.0   1.009061
33     missing_illness   1.004506
```

R: Este modelo controla caracterisicas invariables entre los individuos.

Tenemos un R cuadrado Overall de 0.0137, es decir, el las variables independientes explican un 1,37% la variación en la expectativa de vida y para este modelo en particular nos también destacamos el R cuadrado Within que corresponde al mismo valor. Evidentemente estamos frente a un modelo que explica un bajo porcentaje de la variación de la expectativa de vida, por lo que al igual que el Pooled OLS sugeriría agregar más datos al estudio.

En cuento a la variable educación, el modelo nos entrega que por cada año de educación la expectativa de vida baja en 0,02 años aproximadamente. Si bien, este es un resultado contrainutitivo también es importante destacar que estamos frente a un p muy elevado, por lo que no es significativo.

Otros resultados con un p significativo son, exp que nos dice que mietras más experiencia laboral

tengamos más viviremos y crónica que nos dice que tener una enfermedad crónica disminuye nuestra expectativa de vida. Ambos, resultados esperados.

# 4 Efectos Aleatorios

4. Ejecute un modelo de efectos aleatorios para estimar la misma relacion anterior. Seleccione las variables independientes a incluir en el modelo final e interprete su significado.

```
                        RandomEffects Estimation Summary
================================================================================
Dep. Variable:             expectancy   R-squared:                       0.3537
Estimator:              RandomEffects   R-squared (Between):             0.0533
No. Observations:               54125   R-squared (Within):              0.0045
Date:                Tue, May 20 2025   R-squared (Overall):             0.0381
Time:                        20:53:56   Log-likelihood               -1.894e+05
Cov. Estimator:                Robust
                                        F-statistic:                     1138.6
Entities:                       22898   P-value                          0.0000
Avg Obs:                        2.3637   Distribution:               F(26,54098)
Min Obs:                        1.0000
Max Obs:                        4.0000   F-statistic (robust):            69.007
                                         P-value                          0.0000
Time periods:                        4   Distribution:               F(26,54098)
Avg Obs:                     1.353e+04
Min Obs:                     1.017e+04
Max Obs:                     1.654e+04

                              Parameter Estimates
================================================================================
==
                Parameter  Std. Err.     T-stat    P-value    Lower CI    Upper
CI
--------------------------------------------------------------------------------
--
const              77.690     0.2533     306.69     0.0000      77.193
78.186
edu               -0.0129     0.0104    -1.2436     0.2137     -0.0332
0.0074
kids              -1.5824     0.1061    -14.914     0.0000     -1.7904
-1.3744
exp                0.0364     0.0059     6.2108     0.0000      0.0249
0.0479
cronica           -0.2738     0.1101    -2.4857     0.0129     -0.4897
-0.0579
sistema           -2.2170     0.1314    -16.868     0.0000     -2.4746
-1.9594
wage               0.0860     0.0143     6.0245     0.0000      0.0580
0.1139
```

```
missing_assets    -0.0699     0.1725    -0.4055     0.6851     -0.4080
0.2681
status_4.0         2.7576     0.1785     15.449     0.0000      2.4077
3.1074
informal          -0.6747     0.1827    -3.6925     0.0002     -1.0328
-0.3166
region_3.0        -0.0970     0.4215    -0.2300     0.8181     -0.9231
0.7292
region_4.0         0.6604     0.2759     2.3938     0.0167      0.1197
1.2011
region_5.0         0.5134     0.1836     2.7961     0.0052      0.1535
0.8733
region_6.0         0.7662     0.2318     3.3058     0.0009      0.3119
1.2205
region_8.0        -0.2779     0.1701    -1.6337     0.1023     -0.6114
0.0555
region_9.0        -0.0148     0.2396    -0.0618     0.9507     -0.4845
0.4548
region_11.0       -0.9715     0.6762    -1.4366     0.1508     -2.2970
0.3540
region_13.0        0.6230     0.1299     4.7974     0.0000      0.3685
0.8776
year               0.1470     0.0222     6.6317     0.0000      0.1036
0.1904
situation_4        1.0400     0.1584     6.5648     0.0000      0.7295
1.3505
occupation_2.0     0.7220     0.2264     3.1896     0.0014      0.2783
1.1656
occupation_3.0     0.6907     0.2158     3.2007     0.0014      0.2677
1.1137
occupation_5.0    -0.8878     0.1864    -4.7629     0.0000     -1.2531
-0.5225
occupation_6.0    -0.9673     0.2719    -3.5578     0.0004     -1.5002
-0.4344
occupation_7.0    -0.5173     0.1804    -2.8680     0.0041     -0.8708
-0.1638
occupation_8.0    -0.5612     0.2158    -2.6000     0.0093     -0.9842
-0.1381
occupation_9.0    -0.8691     0.1746    -4.9782     0.0000     -1.2112
-0.5269
==============================================================================
==

Resumen de multicolinealidad:
              Variable        VIF
41         situation_3        inf
29   missing_publicemp        inf
28     missing_selfemp        inf
```

| | | |
|---|---|---|
| 43 | lmp_1 | inf |
| 44 | lmp_2 | inf |
| 42 | situation_4 | 86.076592 |
| 5 | wage | 66.630662 |
| 40 | situation_2 | 33.478850 |
| 27 | missing_hours | 20.214818 |
| 26 | missing_wage | 12.695784 |
| 6 | hours | 10.305009 |
| 25 | missing_informal | 8.434994 |
| 56 | region_13.0 | 6.251521 |
| 34 | missing_assets | 4.621430 |
| 8 | selfemp | 3.938410 |
| 51 | region_8.0 | 3.503090 |
| 0 | year | 3.287589 |
| 48 | region_5.0 | 3.096248 |
| 11 | cotizando | 2.938728 |
| 23 | occupation_9.0 | 2.747175 |
| 50 | region_7.0 | 2.507709 |
| 53 | region_10.0 | 2.464340 |
| 21 | occupation_7.0 | 2.372733 |
| 19 | occupation_5.0 | 2.351576 |
| 49 | region_6.0 | 2.299587 |
| 12 | assets | 2.176963 |
| 52 | region_9.0 | 2.155470 |
| 47 | region_4.0 | 2.007133 |
| 18 | occupation_4.0 | 1.991197 |
| 35 | missing_exp | 1.924878 |
| 22 | occupation_8.0 | 1.924161 |
| 16 | occupation_2.0 | 1.921528 |
| 31 | missing_sistema | 1.853155 |
| 17 | occupation_3.0 | 1.725519 |
| 2 | edu | 1.698473 |
| 20 | occupation_6.0 | 1.617954 |
| 10 | sistema | 1.578722 |
| 1 | hombre | 1.570627 |
| 4 | exp | 1.541304 |
| 45 | region_2.0 | 1.535473 |
| 7 | informal | 1.504890 |
| 39 | status_5.0 | 1.488005 |
| 3 | kids | 1.481231 |
| 15 | occupation_1.0 | 1.399354 |
| 46 | region_3.0 | 1.346611 |
| 13 | cronica | 1.325396 |
| 32 | missing_children | 1.305700 |
| 9 | publicemp | 1.231606 |
| 55 | region_12.0 | 1.201558 |
| 38 | status_4.0 | 1.184901 |
| 54 | region_11.0 | 1.166993 |

```
36          status_2.0   1.136332
57         region_14.0   1.134621
58         region_15.0   1.123802
37          status_3.0   1.113060
30  missing_expectancy   1.085870
14              mental   1.079977
24     occupation_10.0   1.009061
33     missing_illness   1.004506
```

R: Aquí tenemos un R cuadrado de 0,3533, es decir, con este modelo las variables independientes pueden solo pueden explicar un 35,33% de la variación en la expectativa de vida.

En cuanto a la variable de educación, vemos que por cada año de educación la expectativa de vida aumenta en 0,0192 años, además tenemos un p menor a 0,05 por lo que este resultado tiene un buen nivel de significancia.

Otras variables con un p pequeño son, kids que dice que tener hijos disminuye la expectativa de vida y exp que dice que tener más experiencia laboral aumenta la expectativa de vida.

# 5 Comparación de resultados

5. Comente los resultados obtenidos en 2, 3 y 4. ¿Cuáles y por qué existen las diferencias entre los resultados?. En su opinión, ¿Cuál sería el más adecuado para responder la pregunta de investgación y por qué? ¿Qué variables resultaron ser robustas a la especificación?

```
                              Model Comparison
=========================================================================
                                 Pooled             FE                RE
-------------------------------------------------------------------------
Dep. Variable                expectancy     expectancy        expectancy
Estimator                      PanelOLS       PanelOLS     RandomEffects
No. Observations                  54125          54125             54125
Cov. Est.                    Unadjusted         Robust            Robust
R-squared                        0.0601         0.0137            0.3537
R-Squared (Within)               0.0103         0.0137            0.0045
R-Squared (Between)              0.0783        -0.0288            0.0533
R-Squared (Overall)              0.0601        -0.0123            0.0381
F-statistic                      64.063         16.649            1138.6
P-value (F-stat)                 0.0000         0.0000            0.0000
=====================      ============   ============   ===============
const                            78.523         75.010            77.690
                               (127.51)       (75.933)          (306.69)
year                             0.2090         0.1944            0.1470
                               (8.6061)       (6.1612)          (6.6317)
hombre                           1.4776
                               (14.741)
edu                              0.0268        -0.0219           -0.0129
                               (2.6136)      (-0.8476)         (-1.2436)
kids                            -1.5932        -0.2747           -1.5824
```

| | | | |
|---|---|---|---|
| | (-15.940) | (-1.1311) | (-14.914) |
| exp | 0.0151 | 0.0885 | 0.0364 |
| | (2.7210) | (2.3588) | (6.2108) |
| wage | 0.0489 | 0.0476 | 0.0860 |
| | (1.0884) | (2.3031) | (6.0245) |
| hours | -0.0043 | | |
| | (-1.0226) | | |
| informal | -0.2174 | -0.5122 | -0.6747 |
| | (-1.0841) | (-2.0718) | (-3.6925) |
| selfemp | -0.8429 | | |
| | (-3.4406) | | |
| publicemp | -0.3144 | | |
| | (-1.7275) | | |
| sistema | -2.5790 | 0.1199 | -2.2170 |
| | (-15.461) | (0.5546) | (-16.868) |
| cotizando | 1.0895 | | |
| | (7.8352) | | |
| assets | -0.0151 | | |
| | (-1.7108) | | |
| cronica | -0.1136 | -0.6176 | -0.2738 |
| | (-1.0284) | (-3.8492) | (-2.4857) |
| mental | -2.0707 | | |
| | (-12.849) | | |
| occupation_1.0 | -0.3155 | | |
| | (-0.9289) | | |
| occupation_2.0 | 0.4407 | 0.2025 | 0.7220 |
| | (1.5554) | (0.5865) | (3.1896) |
| occupation_3.0 | 0.3463 | 0.4502 | 0.6907 |
| | (1.3088) | (1.5390) | (3.2007) |
| occupation_4.0 | -0.5484 | | |
| | (-2.1812) | | |
| occupation_5.0 | -1.2407 | -0.4727 | -0.8878 |
| | (-5.4493) | (-1.7976) | (-4.7629) |
| occupation_6.0 | -1.5952 | -1.2303 | -0.9673 |
| | (-5.3873) | (-3.1783) | (-3.5578) |
| occupation_7.0 | -1.2085 | -0.3511 | -0.5173 |
| | (-5.4713) | (-1.2654) | (-2.8680) |
| occupation_8.0 | -1.3018 | -0.7188 | -0.5612 |
| | (-5.3232) | (-2.2097) | (-2.6000) |
| occupation_9.0 | -1.3999 | -0.4083 | -0.8691 |
| | (-6.5431) | (-1.5883) | (-4.9782) |
| occupation_10.0 | 2.1459 | | |
| | (1.4506) | | |
| missing_informal | 0.5892 | | |
| | (2.5574) | | |
| missing_wage | 0.6421 | | |
| | (2.2732) | | |
| missing_hours | 0.4589 | | |

| | | | |
|---|---|---|---|
| | (1.3111) | | |
| missing_expectancy | 2.3775 | | |
| | (19.756) | | |
| missing_sistema | 0.7676 | | |
| | (5.5177) | | |
| missing_children | -0.5767 | | |
| | (-4.2710) | | |
| missing_illness | -0.1130 | | |
| | (-0.1062) | | |
| missing_assets | -0.8969 | -0.1183 | -0.0699 |
| | (-4.2220) | (-0.5527) | (-0.4055) |
| missing_exp | -0.7756 | | |
| | (-4.2495) | | |
| status_2.0 | -0.5459 | | |
| | (-3.9566) | | |
| status_3.0 | -0.8301 | | |
| | (-5.5268) | | |
| status_4.0 | 2.7423 | -0.6103 | 2.7576 |
| | (15.020) | (-1.5554) | (15.449) |
| status_5.0 | -0.5700 | | |
| | (-5.1318) | | |
| situation_2 | -2.3241 | | |
| | (-3.9314) | | |
| situation_4 | -1.0262 | -0.2233 | 1.0400 |
| | (-1.7785) | (-1.0568) | (6.5648) |
| region_2.0 | -3.2272 | | |
| | (-9.7713) | | |
| region_3.0 | -1.2084 | 1.8588 | -0.0970 |
| | (-3.1090) | (0.5317) | (-0.2300) |
| region_4.0 | -0.3149 | 0.3352 | 0.6604 |
| | (-1.1315) | (0.1953) | (2.3938) |
| region_5.0 | -0.4834 | 0.4408 | 0.5134 |
| | (-2.0447) | (0.2935) | (2.7961) |
| region_6.0 | -0.2248 | -0.8094 | 0.7662 |
| | (-0.8539) | (-0.5104) | (3.3058) |
| region_7.0 | -0.9736 | | |
| | (-3.8504) | | |
| region_8.0 | -1.6951 | -0.9193 | -0.2779 |
| | (-7.3751) | (-0.4767) | (-1.6337) |
| region_9.0 | -1.1376 | -1.8715 | -0.0148 |
| | (-4.2819) | (-1.0341) | (-0.0618) |
| region_10.0 | -0.9266 | | |
| | (-3.5777) | | |
| region_11.0 | -1.8225 | 2.8322 | -0.9715 |
| | (-3.4548) | (0.5772) | (-1.4366) |
| region_12.0 | -1.0396 | | |
| | (-2.1591) | | |
| region_13.0 | -0.4745 | 1.2083 | 0.6230 |

```
                            (-2.2494)       (1.0036)          (4.7974)
region_14.0                   -0.9405
                            (-1.6515)
region_15.0                   -2.4285
                            (-4.1650)
======================= ============== ============== =================
Effects                                          Entity
-----------------------------------------------------------------------

T-stats reported in parentheses

Hausman Test: chi-2 = 400.208, df = 27, p-value = 4.532401189284201e-68
```

R: Los tres modelos presentan diferentes coeficientes y medidas de ajuste. Estas diferencias se explican principalmente por cómo cada modelo trata la heterogeneidad no observada entre individuos. El modelo de efectos fijos la controla por características no observadas que son constantes en el tiempo, mientras que el modelo de efectos aleatorios asume que hay independencia entre estas variables y las explicativas. El modelo pooled no controla la heterogeneidad, lo que puede producirr sesgos.

Respecto al mejor modelo para la invstigación vemos que el modelo de efectos aleataorios muestra un R cuadrado mucho mayor comparado con efectos fijos y Pooled, sin embargp, esto puede deberse a supuestos más flexibles y no a que el modelo en si tenga mayor validez. Considerando eso el mejor modelo sería el de efectos fijos ya que nos entrega estimaciones más robustas en situacioines donde puede haber correlación entre factores no observados.

En cuanto a las variables más robustas tenemos exp y year que se mantienen postivas en los modelos, y cronica y informal que se mantiene negativa. Edu por otro lado, es muy sensible a los modelos.

Finalmente, el test de Hausman comparó 20 variables y entregó un chicuadrado elevado con un p-valor muy inferior a 0.05, lo que implica que se rechaza fuertemente la hipótesis nula. Esto signfica los efectos no observados están correlacionados con las variables explicativas, y por ello este test nos confirma que el modelo de efectos fijos es el más confiable para este estudio.

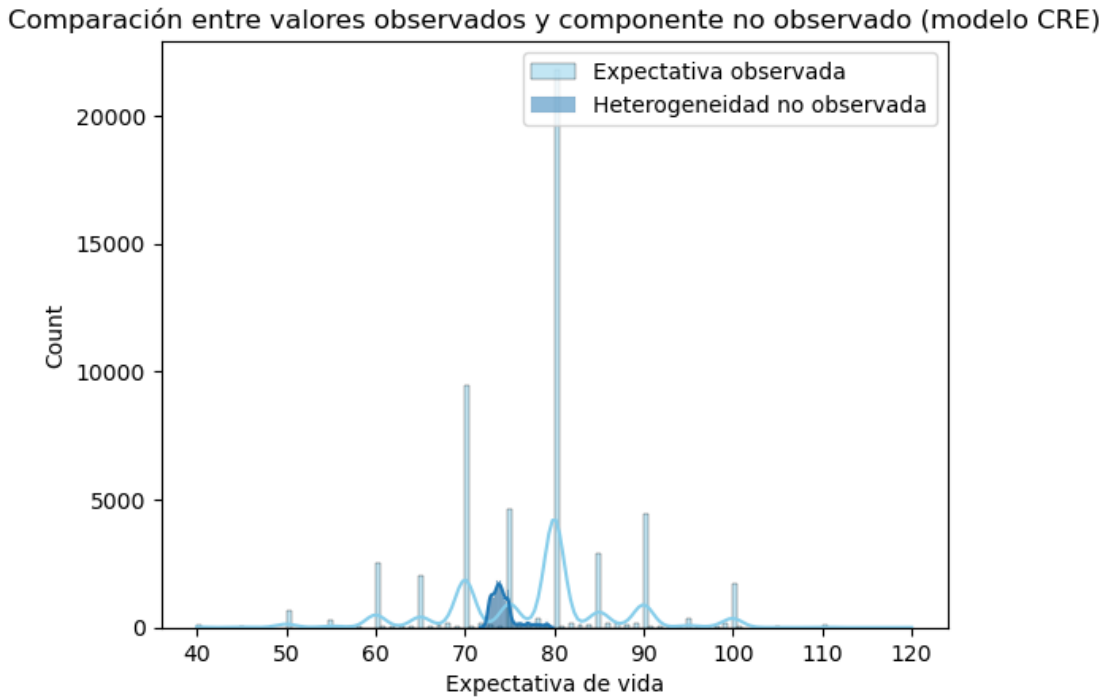
# 6 Efectos Aleatorios Correlacionados

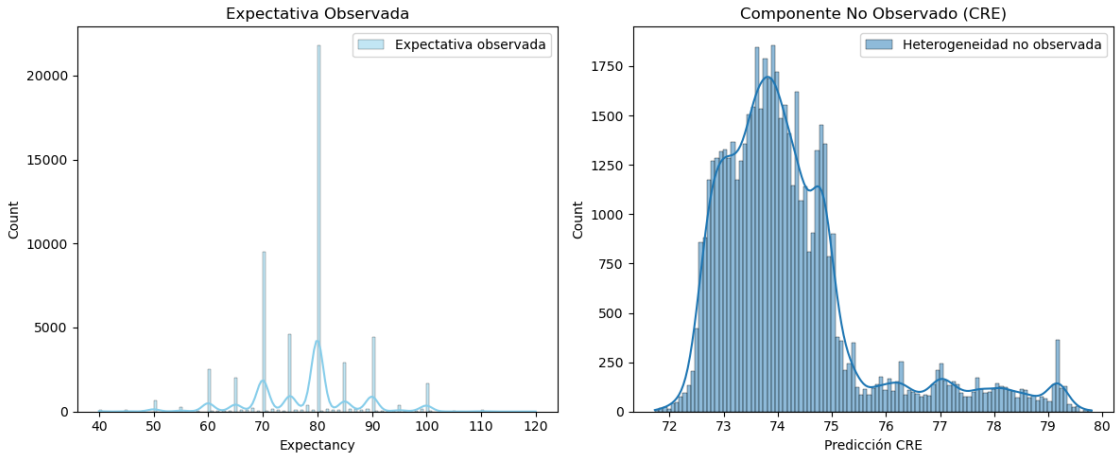
6. Ejecute un modelo de efectos aleatorios correlacionados (CRE) para estimar la misma relacion anterior. Seleccione las variables independientes a incluir en el modelo final e interprete su significado. Es este modelo adecuado, dada la data disponible, para modelar el componente no observado?

```
                        RandomEffects Estimation Summary
==============================================================================
Dep. Variable:             expectancy   R-squared:                       0.3565
Estimator:              RandomEffects   R-squared (Between):             0.0473
No. Observations:               54125   R-squared (Within):              0.0094
Date:                Tue, May 20 2025   R-squared (Overall):             0.0359
Time:                        20:53:57   Log-likelihood               -1.894e+05
Cov. Estimator:                Robust
                                        F-statistic:                     966.49
```

```
Entities:                        22898   P-value                           0.0000
Avg Obs:                        2.3637   Distribution:                F(31,54093)
Min Obs:                        1.0000
Max Obs:                        4.0000   F-statistic (robust):             61.806
                                         P-value                           0.0000
Time periods:                        4   Distribution:                F(31,54093)
Avg Obs:                     1.353e+04
Min Obs:                     1.017e+04
Max Obs:                     1.654e+04

                                Parameter Estimates
================================================================================
===
                 Parameter  Std. Err.     T-stat    P-value    Lower CI    Upper
CI
--------------------------------------------------------------------------------
---
const               79.099     0.2904     272.38     0.0000      78.529
79.668
cronica            -0.2779     0.1109    -2.5066     0.0122     -0.4952
-0.0606
missing_assets     -0.0004     0.1749    -0.0020     0.9984     -0.3431
0.3424
status_4.0          2.7907     0.1784     15.646     0.0000      2.4411
3.1403
edu                -0.0219     0.0232    -0.9435     0.3454     -0.0674
0.0236
exp                 0.0922     0.0286     3.2257     0.0013      0.0362
0.1482
wage                0.1289     0.0170     7.6032     0.0000      0.0957
0.1622
informal           -0.4024     0.2283    -1.7623     0.0780     -0.8499
0.0452
sistema             0.1596     0.2083     0.7661     0.4436     -0.2487
0.5679
region_3.0         -0.3869     0.4241    -0.9122     0.3617     -1.2182
0.4444
region_4.0          0.4826     0.2773     1.7402     0.0818     -0.0610
1.0263
region_5.0          0.2964     0.1848     1.6040     0.1087     -0.0658
0.6586
region_6.0          0.5801     0.2331     2.4889     0.0128      0.1233
1.0369
region_8.0         -0.4819     0.1717    -2.8059     0.0050     -0.8185
-0.1453
region_9.0         -0.1713     0.2412    -0.7100     0.4777     -0.6441
0.3015
region_11.0        -1.1172     0.6790    -1.6455     0.0999     -2.4480
```

```
0.2135
region_13.0         0.4304     0.1318     3.2655     0.0011      0.1720
0.6887
year                0.1529     0.0261     5.8514     0.0000      0.1017
0.2041
situation_4         0.8979     0.1611     5.5733     0.0000      0.5821
1.2137
edu_mean            0.0390     0.0257     1.5211     0.1283     -0.0113
0.0894
exp_mean           -0.0566     0.0292    -1.9362     0.0529     -0.1139
0.0007
wage_mean          -0.0624     0.0176    -3.5547     0.0004     -0.0968
-0.0280
informal_mean      -0.7806     0.3694    -2.1130     0.0346     -1.5047
-0.0565
sistema_mean       -4.3661     0.2666    -16.377     0.0000     -4.8886
-3.8435
occupation_3.0      0.4081     0.2176     1.8755     0.0607     -0.0184
0.8345
occupation_4.0     -0.4005     0.2118    -1.8906     0.0587     -0.8157
0.0147
occupation_5.0     -1.1430     0.1916    -5.9652     0.0000     -1.5186
-0.7674
occupation_6.0     -1.1987     0.2767    -4.3316     0.0000     -1.7410
-0.6563
occupation_7.0     -0.7421     0.1865    -3.9786     0.0001     -1.1077
-0.3765
occupation_8.0     -0.8111     0.2198    -3.6901     0.0002     -1.2418
-0.3803
occupation_9.0     -1.0784     0.1829    -5.8957     0.0000     -1.4370
-0.7199
occupation_10.0     1.3935     1.3026     1.0698     0.2847     -1.1595
3.9466
==============================================================================
===
```




R: Notar que para modelo CRE se incorporaron promedios por individuo de las variables para capturar efectos no observados.

El modelo estimó un R cuadrado de 0,3565 y un R cuadrado overall de 0.0359 con múltiples variables significativas. Por ejemplo, exp y wage son positivos y significativos, mientras que cronica tiene es negativo, lo cual es esperable. Además, los promedios como wage_mean, informal_mean y sistema_mean son significativos, lo que nos valida el uso del modelo de efectos aleatorios.

Los gráficos comparan la distribución observada de la expectativa con la heterogeneidad no observada estimada. En ellos vemos que, al apagar variables observables, el modelo predice, en un rango

mas acotado, una distribución más concentrada, lo que representa el componente no explicado por las variables utilizadas.

Finalmente podemos decir que el modelo CRE es adecuado para este estudio, ya que permite modelar de forma más flexible la heterogeneidad no observada. Además, confirma la robustez de varias variables muy relevantes y es superior al modelo RE.

# 7 Distribucion del componente no observado

7. Usando el modelo CRE, prediga la distribucion del componente no observado. Que puede inferir respecto de la heterogeneidad fija en el tiempo y su impacto en la expectativa de vida?

R: Con el modelo CRE, pudimos predecir la distribución del componente no observado apagando las variables individuales observables y manteniendo los promedios por individuo.

Este enfoque nos permitió aislar la parte de la variación en la expectativa de vida relacionada a la heterogeneidad que es fija a lo largo del timepo. La distribución que obtuvimos es más concentrada y simetrica que la que pudimos observar en los datos reales, es decir, existe un componente individial importante fijo que explica gran parte de los cambios en la expectativa de vida. En conclusión, los factores personales que se mantienen en el tiempo, como genetica, lugar de origen, entre otros, influyen significativamente sobre la expectativa de vida a lo largo del tiempo, lo cual muestra la importancia de modelar correctamente la heterogeneidad no observada.

# 8 Mejor Modelo

8. Usando sus respuestas anteriores, que modelo prefiere? que se puede inferir en general respecto del efecto de las variables explicativas sobre la expectativa de vida?

```
                                    Model Comparison
================================================================================
=========
                                Pooled             FE                RE
CRE
--------------------------------------------------------------------------------
---------
Dep. Variable               expectancy     expectancy        expectancy
expectancy
Estimator                     PanelOLS       PanelOLS     RandomEffects
RandomEffects
No. Observations                 54125          54125            54125
54125
Cov. Est.                   Unadjusted         Robust           Robust
Robust
R-squared                       0.0601         0.0137           0.3537
0.3565
R-Squared (Within)              0.0103         0.0137           0.0045
0.0094
R-Squared (Between)             0.0783        -0.0288           0.0533
0.0473
```

| | | | | |
|---|---|---|---|---|
| R-Squared (Overall) | 0.0601 | -0.0123 | 0.0381 | 0.0359 |
| F-statistic | 64.063 | 16.649 | 1138.6 | 966.49 |
| P-value (F-stat) | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| ========== | ========== | ========== | ========== | ========== |
| const | 78.523 | 75.010 | 77.690 | 79.099 |
| | (127.51) | (75.933) | (306.69) | (272.38) |
| year | 0.2090 | 0.1944 | 0.1470 | 0.1529 |
| | (8.6061) | (6.1612) | (6.6317) | (5.8514) |
| hombre | 1.4776 | | | |
| | (14.741) | | | |
| edu | 0.0268 | -0.0219 | -0.0129 | -0.0219 |
| | (2.6136) | (-0.8476) | (-1.2436) | (-0.9435) |
| kids | -1.5932 | -0.2747 | -1.5824 | |
| | (-15.940) | (-1.1311) | (-14.914) | |
| exp | 0.0151 | 0.0885 | 0.0364 | 0.0922 |
| | (2.7210) | (2.3588) | (6.2108) | (3.2257) |
| wage | 0.0489 | 0.0476 | 0.0860 | 0.1289 |
| | (1.0884) | (2.3031) | (6.0245) | (7.6032) |
| hours | -0.0043 | | | |
| | (-1.0226) | | | |
| informal | -0.2174 | -0.5122 | -0.6747 | -0.4024 |
| | (-1.0841) | (-2.0718) | (-3.6925) | (-1.7623) |
| selfemp | -0.8429 | | | |
| | (-3.4406) | | | |
| publicemp | -0.3144 | | | |
| | (-1.7275) | | | |
| sistema | -2.5790 | 0.1199 | -2.2170 | 0.1596 |
| | (-15.461) | (0.5546) | (-16.868) | (0.7661) |
| cotizando | 1.0895 | | | |
| | (7.8352) | | | |

| | | | | |
|---|---|---|---|---|
| assets | -0.0151 | | | |
| | (-1.7108) | | | |
| cronica | -0.1136 | -0.6176 | -0.2738 | -0.2779 |
| | (-1.0284) | (-3.8492) | (-2.4857) | (-2.5066) |
| mental | -2.0707 | | | |
| | (-12.849) | | | |
| occupation_1.0 | -0.3155 | | | |
| | (-0.9289) | | | |
| occupation_2.0 | 0.4407 | 0.2025 | 0.7220 | |
| | (1.5554) | (0.5865) | (3.1896) | |
| occupation_3.0 | 0.3463 | 0.4502 | 0.6907 | 0.4081 |
| | (1.3088) | (1.5390) | (3.2007) | (1.8755) |
| occupation_4.0 | -0.5484 | | | -0.4005 |
| | (-2.1812) | | | (-1.8906) |
| occupation_5.0 | -1.2407 | -0.4727 | -0.8878 | -1.1430 |
| | (-5.4493) | (-1.7976) | (-4.7629) | (-5.9652) |
| occupation_6.0 | -1.5952 | -1.2303 | -0.9673 | -1.1987 |
| | (-5.3873) | (-3.1783) | (-3.5578) | (-4.3316) |
| occupation_7.0 | -1.2085 | -0.3511 | -0.5173 | -0.7421 |
| | (-5.4713) | (-1.2654) | (-2.8680) | (-3.9786) |
| occupation_8.0 | -1.3018 | -0.7188 | -0.5612 | -0.8111 |
| | (-5.3232) | (-2.2097) | (-2.6000) | (-3.6901) |
| occupation_9.0 | -1.3999 | -0.4083 | -0.8691 | -1.0784 |
| | (-6.5431) | (-1.5883) | (-4.9782) | (-5.8957) |
| occupation_10.0 | 2.1459 | | | 1.3935 |
| | (1.4506) | | | (1.0698) |
| missing_informal | 0.5892 | | | |
| | (2.5574) | | | |
| missing_wage | 0.6421 | | | |
| | (2.2732) | | | |

| | | | | |
|---|---|---|---|---|
| missing_hours | 0.4589 | | | |
| | (1.3111) | | | |
| missing_expectancy | 2.3775 | | | |
| | (19.756) | | | |
| missing_sistema | 0.7676 | | | |
| | (5.5177) | | | |
| missing_children | -0.5767 | | | |
| | (-4.2710) | | | |
| missing_illness | -0.1130 | | | |
| | (-0.1062) | | | |
| missing_assets | -0.8969 | -0.1183 | -0.0699 | -0.0004 |
| | (-4.2220) | (-0.5527) | (-0.4055) | (-0.0020) |
| missing_exp | -0.7756 | | | |
| | (-4.2495) | | | |
| status_2.0 | -0.5459 | | | |
| | (-3.9566) | | | |
| status_3.0 | -0.8301 | | | |
| | (-5.5268) | | | |
| status_4.0 | 2.7423 | -0.6103 | 2.7576 | 2.7907 |
| | (15.020) | (-1.5554) | (15.449) | (15.646) |
| status_5.0 | -0.5700 | | | |
| | (-5.1318) | | | |
| situation_2 | -2.3241 | | | |
| | (-3.9314) | | | |
| situation_4 | -1.0262 | -0.2233 | 1.0400 | 0.8979 |
| | (-1.7785) | (-1.0568) | (6.5648) | (5.5733) |
| region_2.0 | -3.2272 | | | |
| | (-9.7713) | | | |
| region_3.0 | -1.2084 | 1.8588 | -0.0970 | -0.3869 |
| | (-3.1090) | (0.5317) | (-0.2300) | (-0.9122) |
| region_4.0 | -0.3149 | 0.3352 | 0.6604 | 0.4826 |
| | (-1.1315) | (0.1953) | (2.3938) | (1.7402) |
| region_5.0 | -0.4834 | 0.4408 | 0.5134 | 0.2964 |
| | (-2.0447) | (0.2935) | (2.7961) | (1.6040) |
| region_6.0 | -0.2248 | -0.8094 | 0.7662 | 0.5801 |

```
                            (-0.8539)      (-0.5104)          (3.3058)
(2.4889)
region_7.0                    -0.9736
                            (-3.8504)
region_8.0                    -1.6951        -0.9193           -0.2779
-0.4819
                            (-7.3751)      (-0.4767)         (-1.6337)
(-2.8059)
region_9.0                    -1.1376        -1.8715           -0.0148
-0.1713
                            (-4.2819)      (-1.0341)         (-0.0618)
(-0.7100)
region_10.0                   -0.9266
                            (-3.5777)
region_11.0                   -1.8225         2.8322           -0.9715
-1.1172
                            (-3.4548)       (0.5772)         (-1.4366)
(-1.6455)
region_12.0                   -1.0396
                            (-2.1591)
region_13.0                   -0.4745         1.2083            0.6230
0.4304
                            (-2.2494)       (1.0036)          (4.7974)
(3.2655)
region_14.0                   -0.9405
                            (-1.6515)
region_15.0                   -2.4285
                            (-4.1650)
edu_mean
0.0390
 (1.5211)
exp_mean
-0.0566
(-1.9362)
wage_mean
-0.0624
(-3.5547)
informal_mean
-0.7806
(-2.1130)
sistema_mean
-4.3661
(-16.377)
======================= ============== ============== =================
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Effects                                          Entity
-------------------------------------------------------------------------------
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```

```
T-stats reported in parentheses
```

R: Considerando los resultados de los cuatro modelos, el modelo de efectos aleatorios correlacionados es el preferido. Aunque su R cuadrado overall es similar al modelo de efectos aleatorios, el CRE permite modelar la heterogeneidad no observada incluyendo los promedios individuales, lo que es muy relevante y sugerido por el test de Hausman, que rechaza la validez del modelo RE a pesar de su R cuadrado mayor.

Además, el CRE logra estimaciones que son consistentes y significativas para variables relevantes como exp, wage o cronica, mientras que otras variables como sistema_mean e informal_mean son relevantes por su capcacidad de capturar efectos personales o estructurales fijos.

En conslusión, se puede inferir que la expectativa de vida está influida tanto por condiciones actuales (educación, ingresos, salud) como por características individuales que se mantienen fijas en el tiempo, lo que nos dice que la mejor opción para este analisis es el modelo de efectos aleatorios correlacionados.